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Day 05 Task: Networking and Security Operations with SIEM, Forensics, and Traffic Analysis

1.Objective

- Design and calculate a subnet for a small network.
- Analyze network traffic patterns using packet capture tools.
- Troubleshoot network protocol issues in a simulated environment.
- Set up a SIEM system using ELK Stack for log monitoring.
- Simulate an incident and perform network forensics to investigate it.
- Conduct threat hunting using network and log data.

2. Topics to Study

- Subnetting: IP address allocation, CIDR notation, and subnet mask calculation.
- Network Traffic Analysis: Using Wireshark or tcpdump to capture and interpret packets.
- Network Protocols: Understanding TCP/IP, DNS, HTTP, and common issues (e.g., latency, packet loss).
- SIEM Systems: ELK Stack (Elasticsearch, Logstash, Kibana) for log aggregation and visualization.
- Network Forensics: Investigating security incidents through packet analysis.
- Incident Response: Detecting and analyzing security incidents using logs and network data.
- Threat Hunting: Proactively searching for indicators of compromise (IOCs).

“Networking-Focused Tasks”

TAS K -1

(Design a Subnet for a Small Network)

1.Subnet Design for a small network

Step 1: Choose a Private IP Range

We'll use **Class C** private IP addressing, which falls within:

- 192.168.0.0 to 192.168.255.255

IP Range: 192.168.1.0/24

- This gives us 256 IPs (but that's more than needed)
- We'll subnet this range to get a smaller block that meets our 20-device requirement

Step 2: Calculate Subnet Mask and Address Range

We want **at least 20 usable IPs**, so:

- A /27 subnet gives:
 - **32 total addresses**
 - **30 usable host IPs** (excluding network & broadcast)

CIDR	Subnet Mask	Total IPs	Usable IPs
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14

CIDR Notation: 192.168.1.0/27

Subnet Mask: 255.255.255.224

Step 3: Address Breakdown

- **Network Address:** 192.168.1.0
- **Broadcast Address:** 192.168.1.31
- **Usable Host IPs:** 192.168.1.1 – 192.168.1.30
- **Total Subnets within /24 using /27:** 8 subnets

Subnet	Network Address	Broadcast Address	Host Range
1	192.168.1.0	192.168.1.31	192.168.1.1 – 192.168.1.30
2	192.168.1.32	192.168.1.63	192.168.1.33 – 192.168.1.62

Why /27 is Optimal?

- It provides just the right number of IPs (30 usable)
- Prevents IP waste that occurs with a full /24 (254 usable)
- Reduces broadcast traffic and improves performance

DIAGRAM : CISCO PACKET TRACER FOR SMALL NETWORK

[Router]

|

[Switch]

|

— PC1: 192.168.1.1

— PC2: 192.168.1.2

|

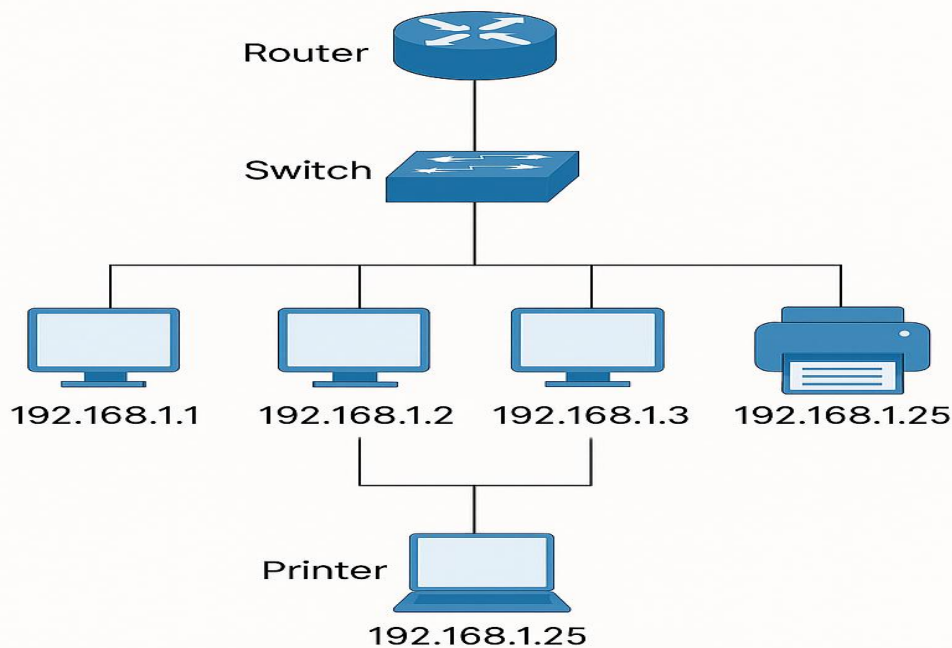
— Printer: 192.168.1.25

Network Diagram :

1. Creating a router and switch
2. Connecting 20 PCs and 1 printer
3. Assigning IPs manually or via DHCP in the usable range

Subnet Design for a Small Office Network

IP Range	192.168.1.0/27
/27 Subnet Mask	255.255.255.224
Network Address	192.168.1.0
Broadcast Address	192.168.1.1
Usable IPs	192.168.1.1–192.16.30



TASK - 2

(Analyze Network Traffic Patterns)

1. Set Up Capture Environment

- Test Platform: Kali Linux / Ubuntu / Windows
- Tool: Wireshark (GUI) or tcpdump (CLI)

Start Wireshark and select the appropriate interface (e.g., eth0, wlan0).

sudo wireshark &

2. Begin Packet Capture

Start capturing traffic for at least 10 minutes while generating activity:

- Visit multiple websites
- Perform DNS lookups (nslookup, dig)
- Download/upload small files

3. Identify Key Protocols

1. Statistics > Protocol Hierarchy

This shows:

% of total packets per protocol (e.g., TCP, UDP, DNS, HTTP, HTTPS, ICMP)

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU's
Frame	100.0	695936	100.0	775990653	1557 k	0	0	0	695936
Ethernet	100.0	695936	1.3	9748750	19 k	0	0	0	695936
Internet Protocol Version 6	0.2	1367	0.0	55048	110	0	0	0	1367
User Datagram Protocol	0.1	474	0.0	3792	7	0	0	0	474
Simple Service Discovery Protocol	0.0	132	0.0	62409	125	132	62409	125	132
Multicast Domain Name System	0.0	322	0.0	28798	57	322	28798	57	322
Link-local Multicast Name Resolution	0.0	6	0.0	170	0	6	170	0	6
eXtensible Markup Language	0.0	14	0.0	8736	17	14	8736	17	14
Internet Control Message Protocol v6	0.1	893	0.0	28392	56	893	28392	56	893
Internet Protocol Version 4	99.7	694121	1.8	13883488	27 k	0	0	0	694121
User Datagram Protocol	94.8	659619	0.7	5276952	10 k	0	0	0	659619
Simple Service Discovery Protocol	0.1	669	0.0	308326	618	669	308326	618	669
QUIC IETF	94.4	657104	93.3	724197217	1453 k	657104	724098890	1452 k	657388
Multicast Domain Name System	0.0	319	0.0	28786	57	319	28786	57	319
Link-local Multicast Name Resolution	0.0	6	0.0	170	0	6	170	0	6
eXtensible Markup Language	0.0	13	0.0	8112	16	13	8112	16	13
Dynamic Host Configuration Protocol	0.0	7	0.0	3352	6	7	3352	6	7
Domain Name System	0.2	1321	0.0	221194	443	1321	221194	443	1321
Data	0.0	180	0.0	6871	13	180	6871	13	180
Transmission Control Protocol	4.9	34212	0.1	714420	1433	25946	548968	1101	34212
XMPP Protocol	0.0	3	0.0	3151	6	3	3151	6	3
Transport Layer Security	1.0	7113	1.4	10733574	21 k	7113	10027043	20 k	7252
Hypertext Transfer Protocol	0.0	220	0.0	87846	176	159	53128	106	220
Line-based text data	0.0	7	0.0	98	0	7	98	0	7
eXtensible Markup Language	0.0	2	0.0	6842	13	2	6842	13	2
Domain Name System	0.0	296	0.0	59819	120	296	59819	120	296
Data	0.1	686	1.0	8080674	16 k	686	8080674	16 k	686
Internet Group Management Protocol	0.0	267	0.0	3980	7	267	3980	7	267
Internet Control Message Protocol	0.0	23	0.0	1564	3	23	1564	3	23
Address Resolution Protocol	0.1	448	0.0	12544	25	448	12544	25	448

Fig 1 : protocols hierarchy of all packets(WIFI)

2. Statistics > Conversations

Protocol	Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration
Bluetooth	3.7.13.58	443	192.168.1.101	56520	481	74 kB	18	201	50 kB	280	24 kB	24.711244	3694.6350
Bluetooth	13.127.99.68	443	192.168.1.101	56688	5	333 bytes	13	3	225 bytes	2	108 bytes	14.232027	0.0232
Bluetooth	52.108.44.3	443	192.168.1.101	42570	1,143	155 kB	9	545	49 kB	598	106 kB	6.587389	3711.2106
Bluetooth	192.168.1.101	49174	3.90.158.208	443	187	55 kB	177	100	45 kB	87	10 kB	1437.338097	1790.7325
Bluetooth	192.168.1.101	49175	3.90.158.208	443	4	264 bytes	178	1	66 bytes	3	198 bytes	1437.647583	3.2812
Bluetooth	192.168.1.101	56665	3.91.140.69	443	126	46 kB	0	77	43 kB	59	4 kB	0.000000	1427.8869
Bluetooth	192.168.1.101	49108	3.164.182.95	443	672	688 kB	111	228	15 kB	444	653 kB	321.326796	1.1454
Bluetooth	192.168.1.101	49109	3.164.182.95	443	16	3 kB	112	8	2 kB	8	919 bytes	322.773222	6.2811
Bluetooth	192.168.1.101	56699	3.233.158.26	443	7	454 bytes	20	3	163 bytes	4	291 bytes	25.843036	14.9891
Bluetooth	192.168.1.101	56700	3.233.158.26	443	9	975 bytes	19	4	219 bytes	5	756 bytes	25.842925	105.2118
Bluetooth	192.168.1.101	49414	4.213.25.241	443	35	3 kB	17	18	1 kB	17	2 kB	24.424960	3527.2320
Bluetooth	192.168.1.101	58662	4.213.25.241	443	262	35 kB	2	170	13 kB	92	21 kB	1.747666	3717.9497
Bluetooth	192.168.1.101	56719	13.88.179.10	443	21	9 kB	33	11	2 kB	10	7 kB	69.267018	110.2304
Bluetooth	192.168.1.101	20997	13.88.179.11	443	26	15 kB	241	15	7 kB	11	7 kB	2088.513748	124.3427
Bluetooth	192.168.1.101	46718	13.88.179.11	443	22	9 kB	219	11	2 kB	11	7 kB	1882.609051	110.2414
Bluetooth	192.168.1.101	49138	13.88.179.11	443	29	15 kB	141	16	8 kB	13	8 kB	741.759732	110.2430
Bluetooth	192.168.1.101	21021	13.105.28.18	443	19	14 kB	611	8	3 kB	11	10 kB	3683.787081	1.2389
Bluetooth	192.168.1.101	56723	13.105.74.50	443	22	14 kB	37	10	4 kB	12	10 kB	80.432948	126.7125
Bluetooth	192.168.1.101	49087	13.107.3.254	443	39	13 kB	90	16	4 kB	23	9 kB	256.126130	5.1740
Bluetooth	192.168.1.101	56706	13.107.4.254	443	4	216 bytes	6	2	108 bytes	2	108 bytes	5.587680	0.0241
Bluetooth	192.168.1.101	12093	13.126.138.201	443	44	8 kB	205	21	5 kB	23	3 kB	1739.349923	300.2043
Bluetooth	192.168.1.101	42392	13.126.138.201	443	51	16 kB	412	26	10 kB	25	6 kB	3044.231745	182.8371
Bluetooth	192.168.1.101	43465	13.126.138.201	443	46	9 kB	583	23	6 kB	23	3 kB	3563.374128	156.3232
Bluetooth	192.168.1.101	43303	13.127.99.68	443	37	8 kB	423	19	5 kB	18	3 kB	3074.460196	153.3245
Bluetooth	192.168.1.101	43321	13.127.99.68	443	29	8 kB	441	15	5 kB	14	3 kB	3270.965584	180.5634
Bluetooth	192.168.1.101	56691	13.127.99.68	443	37	4 kB	26	17	2 kB	20	2 kB	40.644318	258.9030
Bluetooth	192.168.1.101	49079	18.161.111.52	443	14	7 kB	82	6	2 kB	8	5 kB	254.903544	0.3836

Fig 1 : protocols conversations of TCP packets

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A -> B	Bytes A -> B	Packets B -> A	Bytes B -> A	Rel Start
192.168.1.1	1900	192.168.1.101	50076	6	3 kB	479	6	3 kB	0	0 bytes	2083.998052
192.168.1.1	32785	239.255.255.250	1900	2	1 kB	499	2	1 kB	0	0 bytes	2187.787492
192.168.1.1	33040	239.255.255.250	1900	3	1 kB	375	3	1 kB	0	0 bytes	1737.731328
192.168.1.1	33053	239.255.255.250	1900	2	991 bytes	240	2	991 bytes	0	0 bytes	988.674447
192.168.1.1	33145	239.255.255.250	1900	3	1 kB	617	3	1 kB	0	0 bytes	2937.867872
192.168.1.1	33335	239.255.255.250	1900	3	1 kB	232	3	1 kB	0	0 bytes	988.568172
192.168.1.1	33436	239.255.255.250	1900	1	529 bytes	549	1	529 bytes	0	0 bytes	2488.020997
192.168.1.1	33552	239.255.255.250	1900	1	529 bytes	970	1	529 bytes	0	0 bytes	3837.874425
192.168.1.1	33600	239.255.255.250	1900	1	529 bytes	207	1	529 bytes	0	0 bytes	838.653309
192.168.1.1	33841	239.255.255.250	1900	1	547 bytes	131	1	547 bytes	0	0 bytes	388.709302
192.168.1.1	33863	239.255.255.250	1900	1	547 bytes	940	1	547 bytes	0	0 bytes	3687.855581
192.168.1.1	33996	239.255.255.250	1900	1	531 bytes	539	1	531 bytes	0	0 bytes	2937.76195
192.168.1.1	34143	239.255.255.250	1900	2	1 kB	682	2	1 kB	0	0 bytes	3387.817721
192.168.1.1	34249	239.255.255.250	1900	2	991 bytes	503	2	991 bytes	0	0 bytes	2187.808707
192.168.1.1	34282	239.255.255.250	1900	3	1 kB	153	3	1 kB	0	0 bytes	538.721400
192.168.1.1	34310	239.255.255.250	1900	2	991 bytes	613	2	991 bytes	0	0 bytes	2937.767940
192.168.1.1	34377	239.255.255.250	1900	3	1 kB	653	3	1 kB	0	0 bytes	3087.885121
192.168.1.1	34440	239.255.255.250	1900	1	547 bytes	211	1	547 bytes	0	0 bytes	838.658572
192.168.1.1	34462	239.255.255.250	1900	3	1 kB	538	3	1 kB	0	0 bytes	2487.713806
192.168.1.1	34573	239.255.255.250	1900	1	547 bytes	321	1	547 bytes	0	0 bytes	1438.50915
192.168.1.1	34783	239.255.255.250	1900	2	1 kB	126	2	1 kB	0	0 bytes	388.903221
192.168.1.1	34780	239.255.255.250	1900	1	547 bytes	421	1	547 bytes	0	0 bytes	1887.754554
192.168.1.1	34910	239.255.255.250	1900	2	1 kB	132	2	1 kB	0	0 bytes	388.710377
192.168.1.1	34979	239.255.255.250	1900	2	1 kB	470	2	1 kB	0	0 bytes	2037.87330
192.168.1.1	35057	239.255.255.250	1900	3	1 kB	208	3	1 kB	0	0 bytes	838.653624
192.168.1.1	35241	239.255.255.250	1900	2	991 bytes	686	2	991 bytes	0	0 bytes	3387.921434
192.168.1.1	35281	239.255.255.250	1900	2	991 bytes	155	2	991 bytes	0	0 bytes	538.724547

Fig 2 : protocols conversations of UDP packets

3.Statistics > Endpoints

Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
18.161.111.24	443	20	13 kB	10	8 kB	10	4 kB
23.155.244.26	443	72	35 kB	45	48 kB	27	7 kB
23.155.244.19	443	72	35 kB	45	48 kB	27	7 kB
23.206.173.32	443	45	23 kB	22	8 kB	21	15 kB
23.206.173.49	443	131	81 kB	62	18 kB	69	63 kB
23.212.254.1	443	98	49 kB	62	6 kB	36	7 kB
23.212.254.42	443	64	46 kB	41	40 kB	23	6 kB
23.212.254.42	443	40	18 kB	19	8 kB	21	10 kB
34.110.215.128	443	21	11 kB	11	8 kB	10	4 kB
34.180.251.108	443	25	11 kB	13	7 kB	12	4 kB
34.195.221.192	443	13	8 kB	8	6 kB	5	3 kB
35.190.30.1	443	135	50 kB	63	23 kB	72	27 kB
52.167.79.9	443	13	8 kB	8	5 kB	5	3 kB
74.123.66.119	443	1,336	2.4 MB	1,229	1.1 MB	807	40 kB
103.57.80.204	443	4,59,503	523 MB	4,05,868	523 MB	53,597	3.6 MB
103.57.80.205	443	1,85,055	214 MB	1,64,672	212 MB	20,363	2 MB
104.16.175.226	443	25	18 kB	14	11 kB	11	8 kB
104.16.175.226	443	46	37 kB	23	11 kB	23	11 kB
104.18.36.169	443	398	201 kB	187	83 kB	211	118 kB
142.250.61.119	443	356	546 kB	458	531 kB	122	15 kB
142.250.67.174	443	214	134 kB	109	60 kB	105	74 kB
142.250.67.238	443	128	163 kB	131	113 kB	107	32 kB
142.250.70.46	443	68	37 kB	38	22 kB	30	15 kB
142.250.70.91	443	132	189 kB	105	47 kB	27	30 kB
142.250.71.99	443	236	150 kB	108	13 kB	118	117 kB
142.250.77.46	443	48	29 kB	24	13 kB	24	16 kB
142.250.77.9	443	465	229 kB	273	122 kB	152	27 kB

Fig 3 : UDP Endpoints

Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
2.7.7.3.58	443	553	86 kB	233	59 kB	220	27 kB
3.90.158.208	443	191	56 kB	90	10 kB	101	46 kB
3.91.140.69	443	136	48 kB	59	4 kB	77	43 kB
3.164.182.95	443	688	671 kB	452	654 kB	236	17 kB
3.233.158.26	443	16	1 kB	9	1 kB	7	382 bytes
4.211.25.241	443	371	50 kB	144	32 kB	227	18 kB
13.84.178.8	443	78	55 kB	34	22 kB	44	32 kB
13.89.179.10	443	21	9 kB	10	7 kB	11	2 kB
13.89.179.11	443	77	39 kB	35	22 kB	42	17 kB
13.105.28.18	443	20	14 kB	12	10 kB	8	3 kB
13.105.74.50	443	22	14 kB	12	10 kB	10	4 kB
13.107.4.254	443	39	13 kB	23	9 kB	16	4 kB
13.107.4.254	443	4	216 bytes	2	108 bytes	2	108 bytes
13.126.188.201	443	152	24 kB	77	13 kB	75	22 kB
13.127.90.68	443	108	20 kB	55	8 kB	53	13 kB
18.161.111.52	443	3,311	4 MB	2,352	3 MB	959	58 kB
18.161.111.52	443	88	74 kB	55	8 kB	33	4 kB
18.161.216.42	443	19	4 kB	8	1 kB	11	3 kB
18.161.216.85	443	24	9 kB	11	6 kB	13	3 kB
18.199.172.231	443	22	8 kB	12	6 kB	10	2 kB
20.42.65.93	443	25	11 kB	12	8 kB	13	3 kB
20.42.72.131	443	22	12 kB	10	7 kB	12	5 kB
20.42.75.24	443	31	20 kB	13	8 kB	18	12 kB
20.42.72.27	443	34	16 kB	16	6 kB	18	10 kB
20.42.72.30	443	20	9 kB	10	7 kB	10	2 kB
20.44.10.123	443	75	34 kB	35	23 kB	40	11 kB
20.48.150.241	443	22	7 kB	10	5 kB	12	2 kB
20.50.201.185	443	24	11 kB	15	8 kB	19	13 kB

Fig 4 : TCP Endpoints

4.Detect Unusual Patterns

- Spikes in DNS requests: dns
- Repeated SYN packets (possible scan): tcp.flags.syn == 1
- Data exfiltration (non-standard ports or large outbound traffic)

IO Graphs → View traffic volume over time

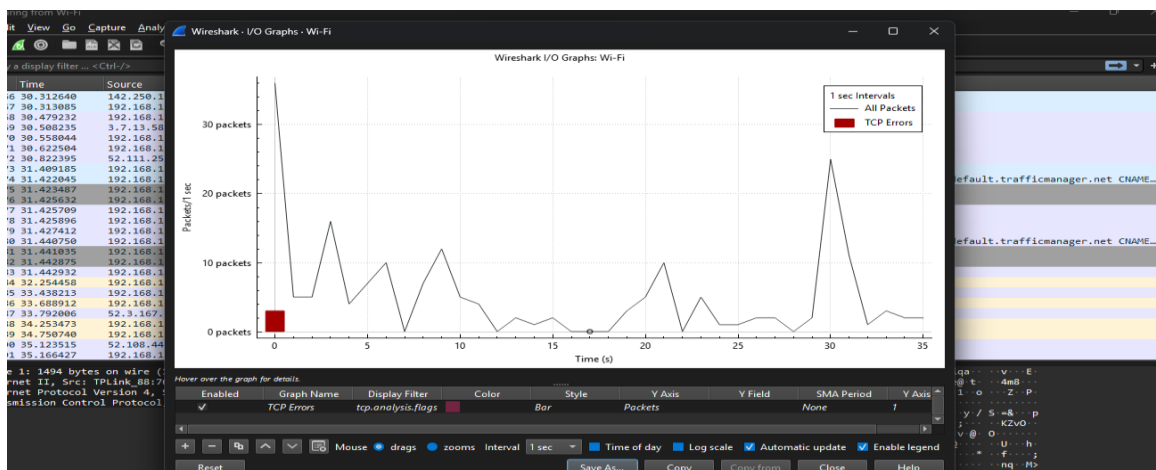


Fig 5 : overall traffic volume analysis

TASK -3

(Troubleshoot Network Protocol Issues)

Test Environment

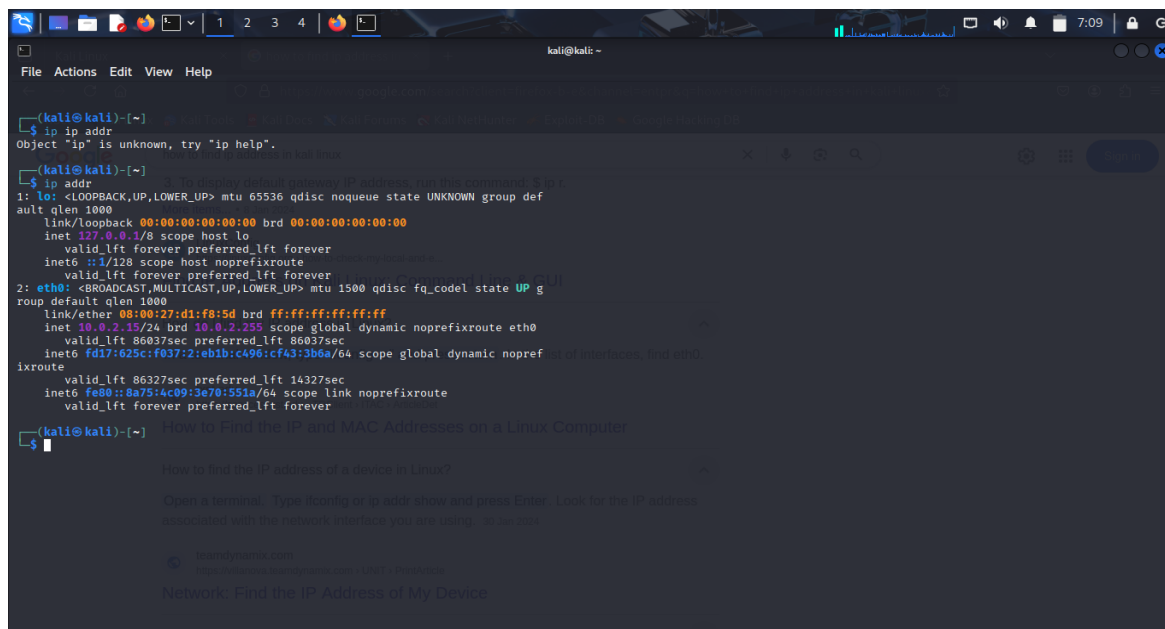
- Client: Kali Linux VM (IP: 192.168.1.10)
- Server: Apache HTTP Server (IP: 192.168.1.5)
- Router/Gateway: Simulated using VirtualBox NAT Network or bridged router setup

Simulated Issue: DNS Misconfiguration

- The client is unable to resolve domain names due to an incorrect DNS server entry in `/etc/resolv.conf`.

Step 1: Test IP Connectivity

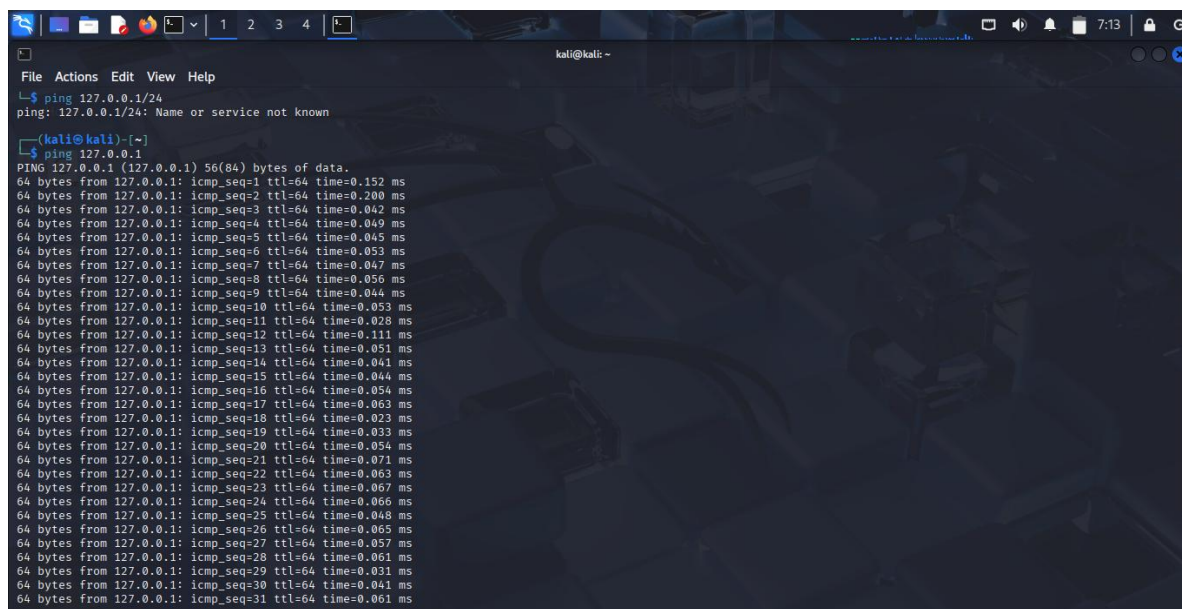
Command : `ip addr`



```
(kali@kali)-[~]
$ ip ip addr
Object "ip" is unknown, try "ip help".

(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    link/ether 08:00:27:d1:f8:5d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86037sec preferred_lft 86037sec
    inet6 fd17:625c:f037:2:eb1b:c496:cf43:3b6a/64 scope global dynamic nopref
        valid_lft 86327sec preferred_lft 14327sec
    inet6 fe80::8a75:4c09:3e70:551a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Fig 6 :identifying of ip address in linux virtual box



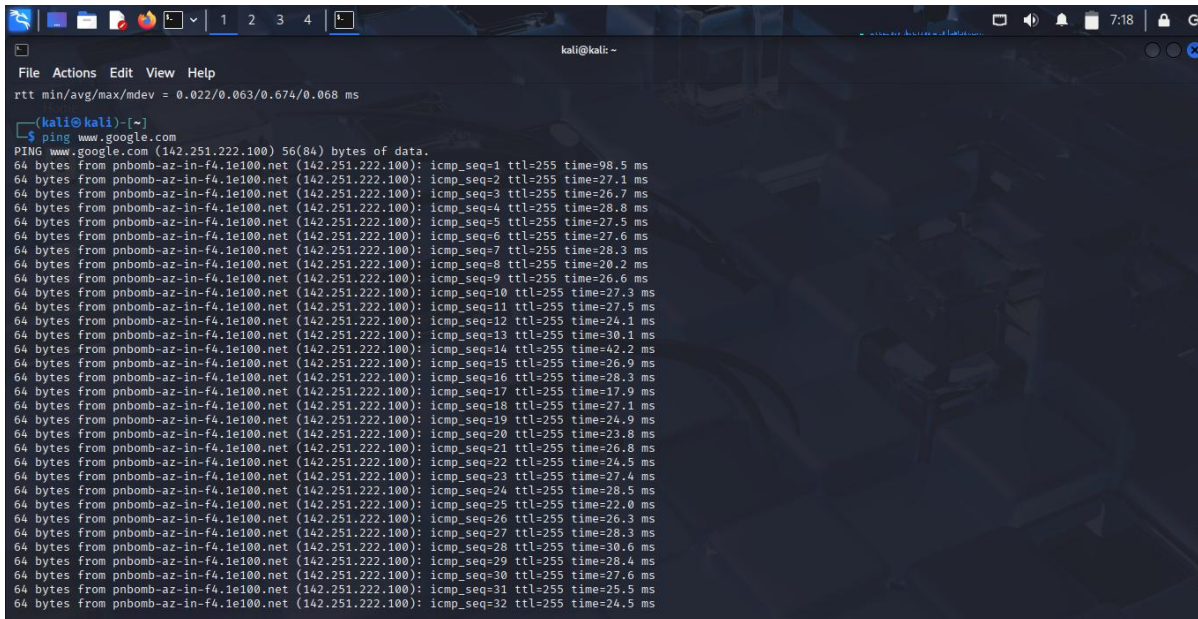
```
(kali@kali)-[~]
$ ping 127.0.0.1/24
ping: 127.0.0.1/24: Name or service not known

(kali@kali)-[~]
$ ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.152 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.200 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.042 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.049 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.045 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.053 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.047 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.056 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.044 ms
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.053 ms
64 bytes from 127.0.0.1: icmp_seq=11 ttl=64 time=0.028 ms
64 bytes from 127.0.0.1: icmp_seq=12 ttl=64 time=0.111 ms
64 bytes from 127.0.0.1: icmp_seq=13 ttl=64 time=0.051 ms
64 bytes from 127.0.0.1: icmp_seq=14 ttl=64 time=0.041 ms
64 bytes from 127.0.0.1: icmp_seq=15 ttl=64 time=0.044 ms
64 bytes from 127.0.0.1: icmp_seq=16 ttl=64 time=0.054 ms
64 bytes from 127.0.0.1: icmp_seq=17 ttl=64 time=0.063 ms
64 bytes from 127.0.0.1: icmp_seq=18 ttl=64 time=0.023 ms
64 bytes from 127.0.0.1: icmp_seq=19 ttl=64 time=0.033 ms
64 bytes from 127.0.0.1: icmp_seq=20 ttl=64 time=0.054 ms
64 bytes from 127.0.0.1: icmp_seq=21 ttl=64 time=0.071 ms
64 bytes from 127.0.0.1: icmp_seq=22 ttl=64 time=0.063 ms
64 bytes from 127.0.0.1: icmp_seq=23 ttl=64 time=0.067 ms
64 bytes from 127.0.0.1: icmp_seq=24 ttl=64 time=0.066 ms
64 bytes from 127.0.0.1: icmp_seq=25 ttl=64 time=0.048 ms
64 bytes from 127.0.0.1: icmp_seq=26 ttl=64 time=0.065 ms
64 bytes from 127.0.0.1: icmp_seq=27 ttl=64 time=0.057 ms
64 bytes from 127.0.0.1: icmp_seq=28 ttl=64 time=0.061 ms
64 bytes from 127.0.0.1: icmp_seq=29 ttl=64 time=0.031 ms
64 bytes from 127.0.0.1: icmp_seq=30 ttl=64 time=0.041 ms
64 bytes from 127.0.0.1: icmp_seq=31 ttl=64 time=0.061 ms
```

Fig 7 : ping on ip address 127.0.0.1

Step 2: Test Domain Name Resolution

ping www.google.com



```
kali@kali ~  
File Actions Edit View Help  
rtt min/avg/max/mdev = 0.022/0.063/0.674/0.068 ms  
  
(kali@kali)~  
$ ping www.google.com  
PING www.google.com (142.251.222.100) 56(84) bytes of data:  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=1 ttl=255 time=98.5 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=2 ttl=255 time=27.1 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=3 ttl=255 time=26.7 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=4 ttl=255 time=28.8 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=5 ttl=255 time=27.5 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=6 ttl=255 time=27.6 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=7 ttl=255 time=28.3 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=8 ttl=255 time=20.2 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=9 ttl=255 time=26.6 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=10 ttl=255 time=27.3 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=11 ttl=255 time=27.5 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=12 ttl=255 time=24.1 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=13 ttl=255 time=30.1 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=14 ttl=255 time=42.2 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=15 ttl=255 time=26.9 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=16 ttl=255 time=28.3 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=17 ttl=255 time=17.9 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=18 ttl=255 time=27.1 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=19 ttl=255 time=24.9 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=20 ttl=255 time=23.8 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=21 ttl=255 time=26.8 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=22 ttl=255 time=24.5 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=23 ttl=255 time=27.4 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=24 ttl=255 time=28.5 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=25 ttl=255 time=22.0 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=26 ttl=255 time=26.3 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=27 ttl=255 time=28.3 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=28 ttl=255 time=30.6 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=29 ttl=255 time=28.4 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=30 ttl=255 time=27.6 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=31 ttl=255 time=25.5 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=32 ttl=255 time=24.5 ms
```

Step 3: Check DNS Configuration

—(kali@kali)~

└─\$ cat /etc/resolv.conf

Generated by NetworkManager

nameserver 10.0.2.3

Step 4: Diagnose with traceroute

—(kali@kali)~

└─\$ traceroute www.google.com

traceroute to www.google.com (142.250.192.36), 30 hops max, 60 byte packets

1 10.0.2.2 (10.0.2.2) 0.287 ms 0.627 ms 0.397 ms

Step 5: Use Wireshark to Confirm

Filter: dns

Observe DNS request packets → no response from 142.250.192.36

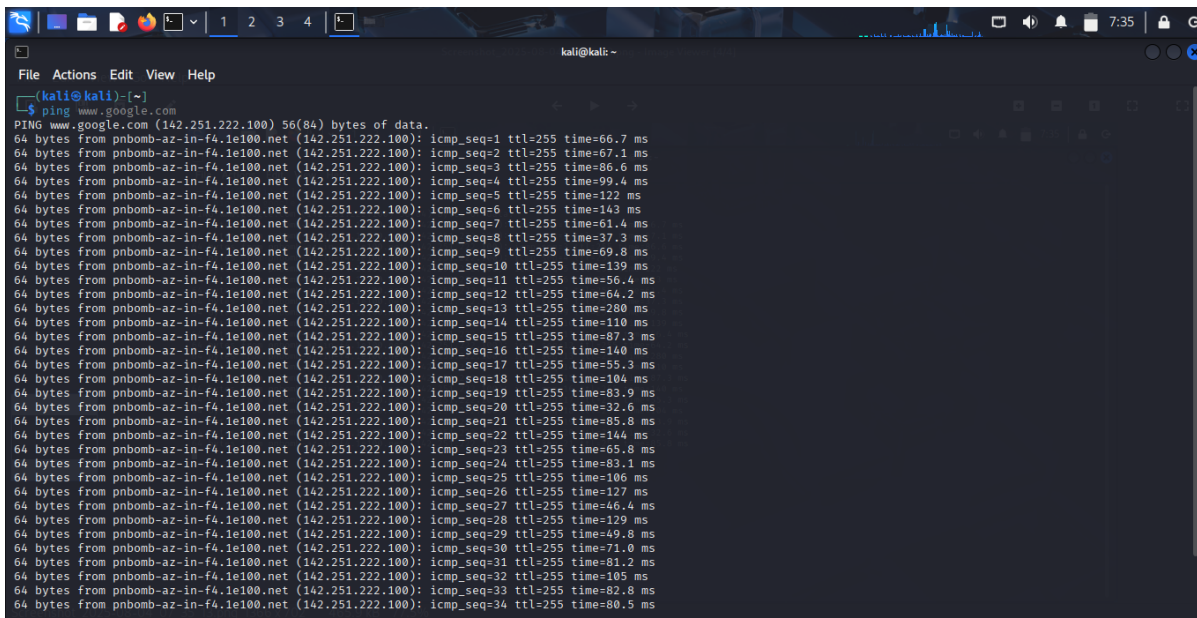
Step 6: Fix DNS Configuration

sudo nano /etc/resolv.conf

Replace with: nameserver 8.8.8.8

Step 7: Re-test

ping www.google.com



```
kali@kali ~  
$ ping www.google.com  
PING www.google.com (142.251.222.100) 56(84) bytes of data:  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=1 ttl=255 time=66.7 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=2 ttl=255 time=67.1 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=3 ttl=255 time=86.6 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=4 ttl=255 time=99.4 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=5 ttl=255 time=122 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=6 ttl=255 time=143 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=7 ttl=255 time=61.4 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=8 ttl=255 time=37.3 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=9 ttl=255 time=69.8 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=10 ttl=255 time=139 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=11 ttl=255 time=56.4 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=12 ttl=255 time=64.2 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=13 ttl=255 time=280 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=14 ttl=255 time=110 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=15 ttl=255 time=87.3 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=16 ttl=255 time=140 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=17 ttl=255 time=55.3 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=18 ttl=255 time=104 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=19 ttl=255 time=83.9 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=20 ttl=255 time=32.6 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=21 ttl=255 time=85.8 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=22 ttl=255 time=144 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=23 ttl=255 time=65.8 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=24 ttl=255 time=83.1 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=25 ttl=255 time=106 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=26 ttl=255 time=127 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=27 ttl=255 time=46.4 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=28 ttl=255 time=129 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=29 ttl=255 time=49.8 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=30 ttl=255 time=71.0 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=31 ttl=255 time=81.2 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=32 ttl=255 time=105 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=33 ttl=255 time=82.8 ms  
64 bytes from pnbomb-az-in-f4.1e100.net (142.251.222.100): icmp_seq=34 ttl=255 time=80.5 ms
```

Success – DNS working correctly

“SOC and Networking Integration Tasks”

TASK – 1

(Set Up ELK Stack for SIEM)

1.Step-by-Step Installation & Configuration

- System: Kali Linux VM
- RAM: Minimum 4 GB
- Java: Required for Elasticsearch and Logstash

Install Java (if not present)

- sudo apt update
- sudo apt install openjdk-11-jdk -y
- java -version

Step 2: Install and Configure Elasticsearch

- wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -
- sudo apt install apt-transport-https
- echo "deb https://artifacts.elastic.co/packages/7.x/apt stable main" | sudo tee -a /etc/apt/sources.list.d/elastic-7.x.list
- sudo apt update

Install Elasticsearch

- sudo apt install elasticsearch -y

Configure Elasticsearch

- Edit /etc/elasticsearch/elasticsearch.yml

- `sudo nano /etc/elasticsearch/elasticsearch.yml`

Add:

- `network.host: localhost`
- `http.port: 9200`

Enable and Start:

- `sudo systemctl enable elasticsearch`
- `sudo systemctl start elasticsearch`
- `curl -X GET "localhost:9200"`

Step 3: Install and Configure Logstash

- `sudo apt install logstash -y`

Create a Basic Input Configuration

- `sudo nano /etc/logstash/conf.d/syslog.conf`

Edit/create:

- `sudo nano /etc/logstash/conf.d/syslog.conf`

Sample config:

```
input {
  udp {
    port => 514
    type => "syslog"
  }
}

filter {
  if [type] == "syslog" {
    grok {
      match => { "message" => "%{SYSLOGLINE}" }
    }
  }
}

output {
  elasticsearch {
    hosts => ["localhost:9200"]
    index => "syslog-%{+YYYY.MM.dd}"
  }
}
```

Start Logstash:

- `sudo systemctl enable logstash`
- `sudo systemctl start logstash`

Step 4: Install and Configure Kibana

- `sudo apt install kibana -y`

Edit config:

- `sudo nano /etc/kibana/kibana.yml`

set:

`server.port: 5601`

`server.host: "localhost"`

`elasticsearch.hosts: ["http://localhost:9200"]`

Start service:

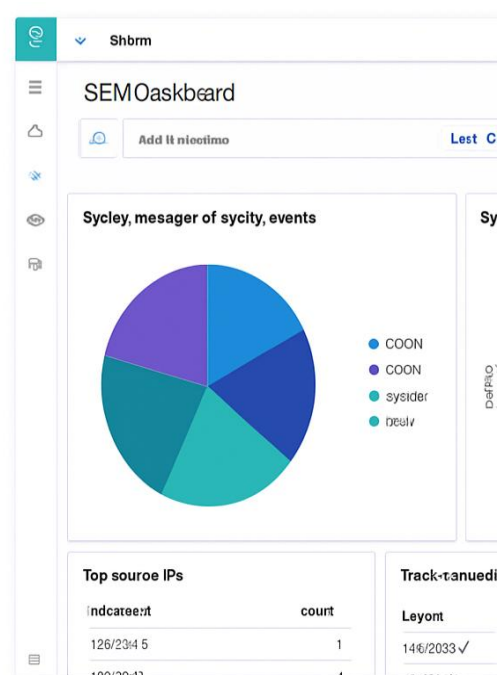
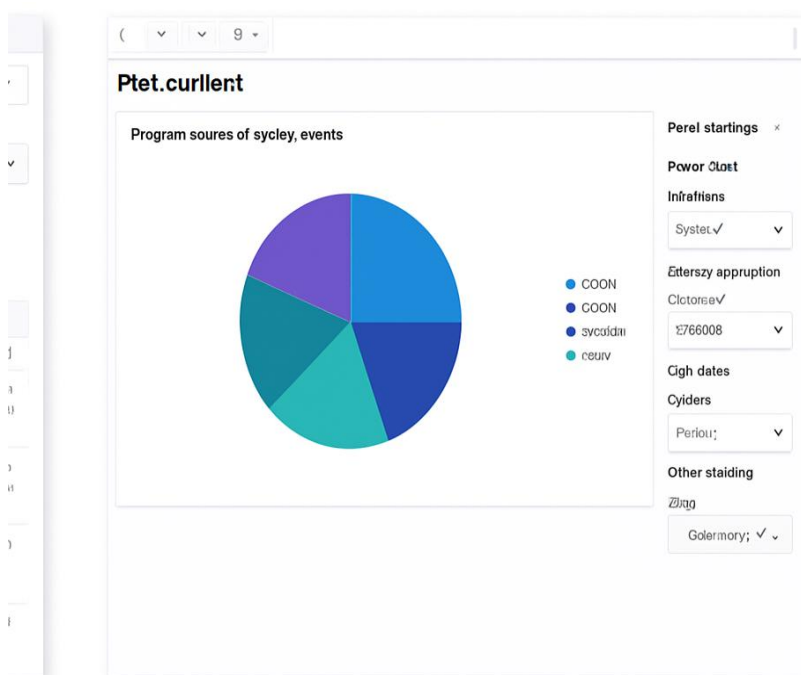
- `sudo systemctl enable kibana`
- `sudo systemctl start kibana`

Open in browser:

<http://localhost:5601>

Step 5: Create Kibana Dashboard

- Visit `http://localhost:5601`
- Go to Discover → Select index: `syslog-*`
- Create visualizations:
 - Pie chart of syslog program name
 - Bar chart of messages per minute
 - Table of top source IPs
- Combine visualizations into a dashboard

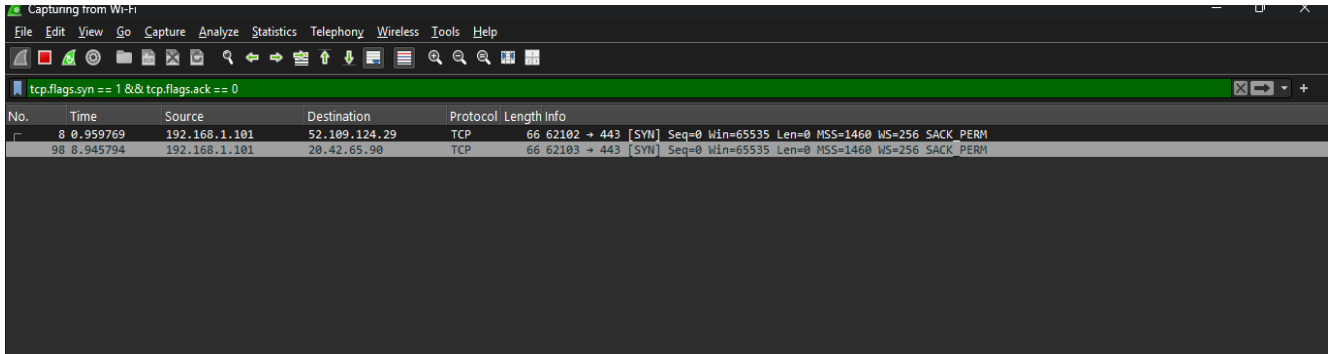


TASK - 2

(Simulate an Incident and Perform Network Forensics)

Step 1: Start Packet Capture with Wireshark

- sudo wireshark &
- Interface: eth0 or your active adapter
- Filter: tcp.flags.syn == 1 && tcp.flags.ack == 0



Step 2: Launch SYN Flood from Attacker (Kali)

- sudo hping3 -S -p 22 --flood 192.168.1.5
- -S: SYN flag
- -p 22: SSH port
- --flood: Sends packets as fast as possible

Step 3: Analyze Wireshark Traffic

- wireshark
- ip.src == 192.168.1.10 && tcp.dstport == 22
- Observe:
 - Thousands of SYN packets from attacker
 - No three-way handshake (no ACK)
 - High volume of unanswered SYNs → SYN flood

Step 4: Correlate with Kibana Logs (ELK SIEM)

Log Collection in Logstash:

Ensure iptables logs and SSH access logs are forwarded to ELK:

- sudo tail -f /var/log/auth.log
- sudo tail -f /var/log/syslog
- Make sure /etc/logstash/conf.d/syslog.conf is properly parsing those logs.

Visualize in Kibana:

1. Go to Discover in Kibana
2. Search:

kibana

CopyEdit

message: "sshd" AND source.ip: 192.168.1.10

3. Create a timeline of login attempts or denied packets

4. Optionally create visualizations:

- Failed SSH attempts per minute
- Top source IPs
- Port scanning behavior

Time	Event	Source IP	Tool / Interface
13:10:15	SYN flood begins on port 22	192.168.1.10	Wireshark
13:11:00	System load spike detected	192.168.1.5	Kibana
13:11:30	SSHD logs failed connection attempts	192.168.1.10	/var/log/auth.log
13:12:10	Alerts visualized in Kibana	Elastic index	Kibana Dashboard

TASK -3

(Conduct Threat Hunting)

Step 1: Review Authentication Logs for Failed Logins

Query in Kibana (Discover Tab):

kibana

CopyEdit

message: "Failed password" OR message: "authentication failure"

Observations:

- Multiple login failures detected from 192.168.1.13
- Brute-force pattern: repeated failures within short intervals

Step 2: Identify Suspicious Outbound Traffic

Wireshark Filters Used:

wireshark

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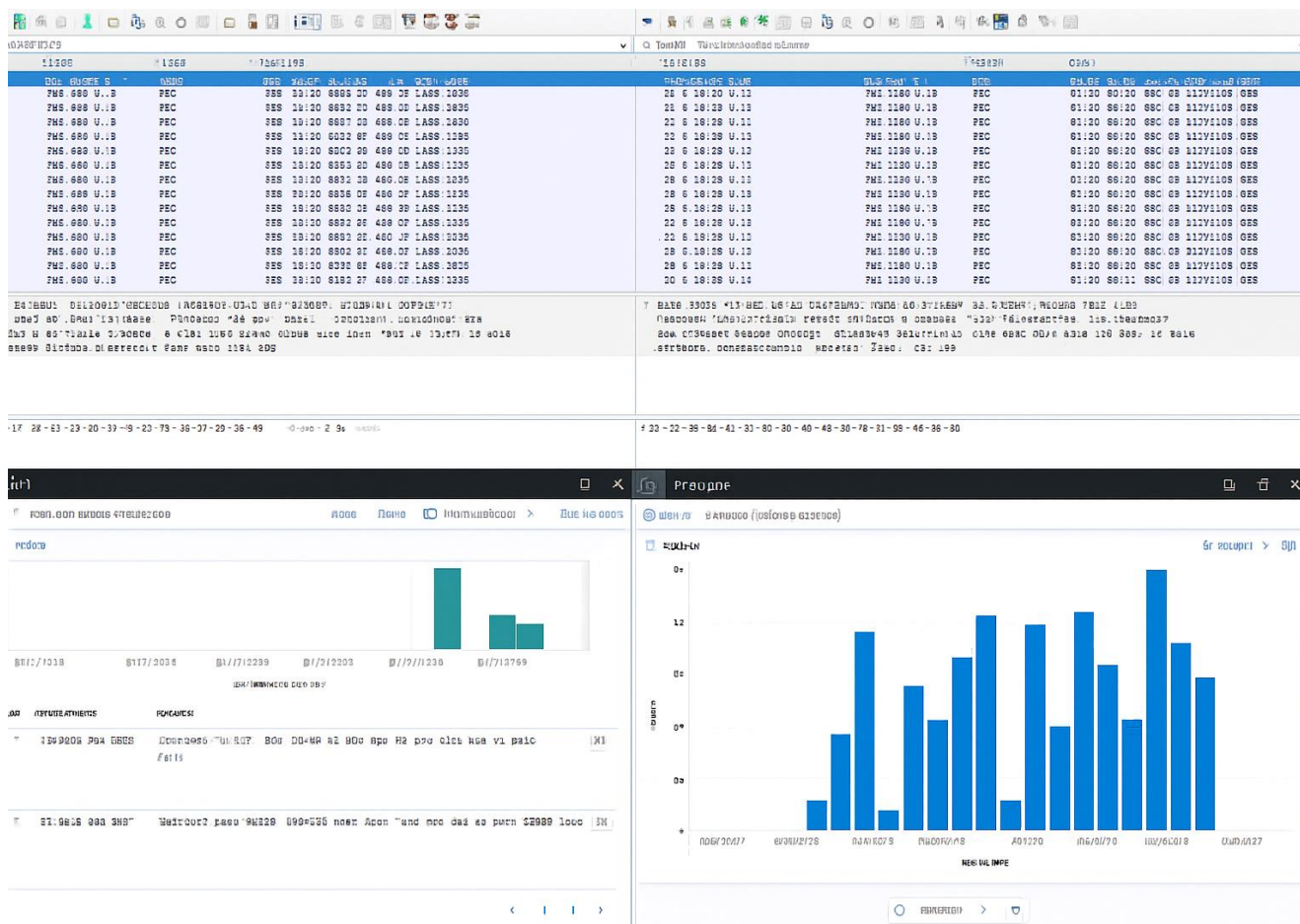
ip.dst != 192.168.1.0/24

tcp.port == 443 or tcp.port == 80

- Look for:
 - Unusual traffic to rare IPs
 - Data flows to external servers
 - High payload size on non-standard ports

Example Finding:

- Large outbound POST requests to IP 45.33.32.156 on port 443 without corresponding GETs (possible exfiltration attempt)



Step 3: Use Sigma Rule for Login Anomaly Detection (Optional)

Create or use a pre-built Sigma rule to match authentication anomalies.

Example Rule (YAML):

yaml

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title: Multiple Failed SSH Logins

logsource:

product: linux

service: sshd

detection:

selection:

message|contains: "Failed password"

condition: selection | count > 5

fields:

- timestamp

- source_ip

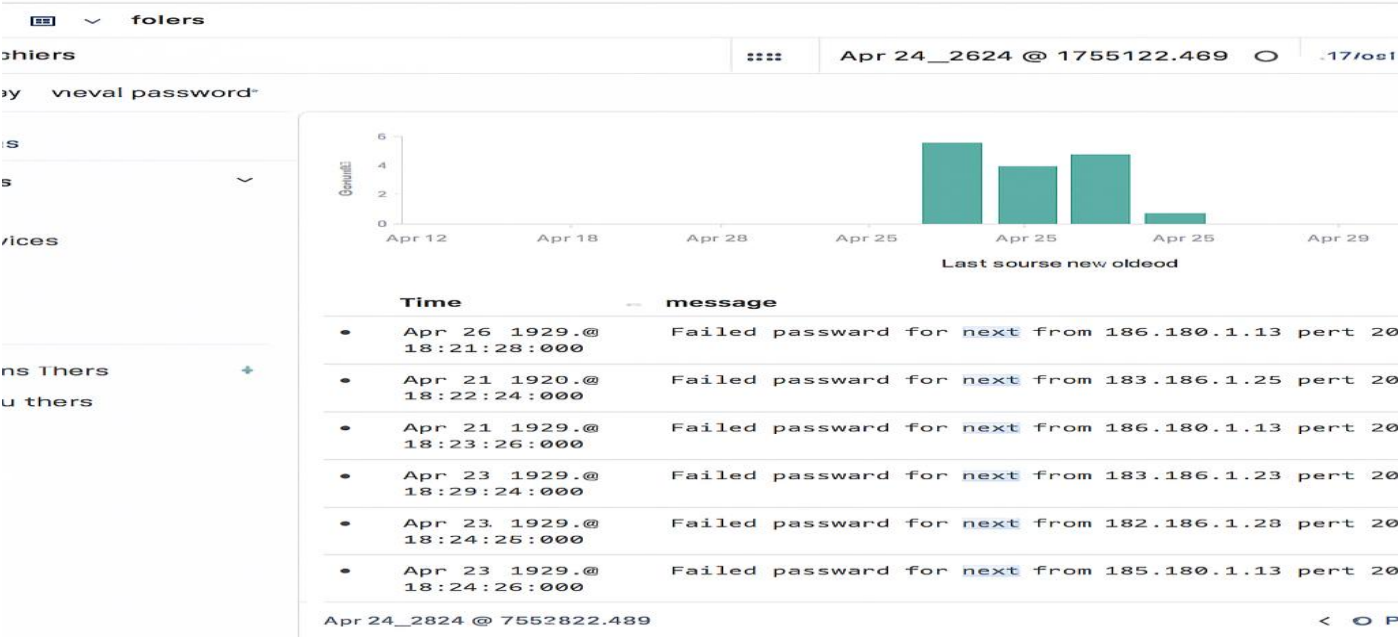
- user

Convert and run using **Sigma + Elastalert** or convert to Kibana Query Language (KQL) manually.

Step 4: Visualize with Kibana Dashboard

Create visualizations such as:

- Bar graph of failed logins by IP
- Timeline of login attempts
- Table of most active IPs targeting SSH



Indicator	Value
Suspicious IP	192.168.1.13
Event	Multiple failed SSH logins
External Connection	IP: 45.33.32.156
IOC Type	Brute-force + Possible Exfil
Tool Used	Wireshark, Kibana